Nobuharu Shimazu

Seattle, WA · nobu.bichanna@gmail.com · 571-241-8705 · github.com/bichanna · linkedin.com/in/nobuharu-shimazu

Systems and programming-languages developer focused on language design, virtual machines, runtimes, and developer tooling—author of a published Rust crate, a systems language compiler, and a cross-platform language interpreter.

EDUCATION

University of Washington — Seattle, WA Expected 2029

B.S. in Computer Science, Paul G. Allen School of Computer Science & Engineering

Relevant Coursework: CSE 122–123 (Programming & Data Structures), MATH 208 (Linear Algebra), MATH 124–125 (Calculus)

EXPERIENCE

Mazu Bay — Co-founder & Lead Engineer 2026–Present

Independent two-person game studio — C / Godot 4 / GDScript

- Co-founded the studio and lead engine and gameplay programming.
- Engineered core systems in Godot 4: a combat state machine, modular weapon system, reusable hitbox/hurtbox component architecture, and behavior-tree enemy AI (LimboAI).
- Designing and building two in-development titles: Sinborn, a first-person dark-fantasy Soulslike boss-rush, and SECOND, a boxing management game.

Daedalus Robotics — Software Engineer 2022–2024

- Software engineer on a back-to-back national champion team in the Bell Advanced Vertical Robotics (AVR) competition.
- Built autonomy software, flight controls, and simple computer-vision pipelines for an autonomous quadcopter in a ROS 2-based stack.

PROJECTS

Dray — Reference-counted systems language & compiler

- Designing and implementing a reference-counted systems language compiling to C, with a Go-style grammar, explicit-receiver methods, and duck-typed generics via comptime.
- Built the compiler pipeline (lexing through codegen) in Rust on top of Tamago, my own C-emission crate.

Tamago — Open-source Rust crate

- Authored and published a Rust library to crates.io (~1,000 downloads) that programmatically emits formatted, compilable C through a typed builder API.

Crave — Recipe vault/builder native desktop application

- Built a native, cross-platform recipe manager in C17 with Clay immediate-mode UI, Raylib rendering, SQLite storage; structured with The Elm Architecture (TEA) across a modular, multi-file codebase.
- Shipped v0.1.0 with automated cross-platform release builds via GitHub Actions; usable end-to-end, with a backlog of planned features.

SLAP — Programming language & interpreter

- Built a tree-walking interpreter for a dynamically- and strongly-typed, object-oriented language with C-family syntax (implemented in Nim).
- Configured simple cross-platform CI (Linux/macOS/Windows) and documentation. Built a SLAP-to-JavaScript transpiler.

Darcie — Bytecode virtual machine

- Built a bytecode virtual machine in C for a dynamically typed language, with first-class closures and an integrated garbage collector.
- Implemented a primitive C foreign-function interface (FFI) for calling native code from the language.

FLan — Functional-language runtime

- Implemented a generational garbage collector in C for an experimental functional language, automating memory reclamation across object generations.

Additional: rayanim, a Manim-inspired animation engine with Raylib and C; a prototype bytecode VM in C++.

WRITING

bichanna.github.io/posts — technical blog on language design, compilers, and systems programming.

TECHNICAL SKILLS

Languages: C, Rust, Java, Go, Python
Tools & Technologies: Git, ROS 2, SQLite, Raylib, Clay, Godot 4, Django, Meson/Ninja
Spoken Languages: Japanese, English, Vietnamese